Week 11 - Introduction to Modeling, part 2

1. Lesson - No lesson this week

2. Weekly graph question

In [6]:
```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import root_mean_squared_error
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import make_scorer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import RepeatedKFold
from sklearn.model_selection import cross_val_score
from sklearn.metrics import classification_report
```

The book names one of Vonnegut's rules as "keep it simple" and another as "have the guts to cut." Here is some data from the previous week's lesson. If you had to cut one of the two plots below, which would it be? Which seems more interesting or important? Explain. (Should "amount of training data used" or "number of estimators" be on the x-axis.)

In [7]:
```
np.random.seed(0)
num_points = 10000
feature_1a = np.random.random(size = num_points) * 3
feature_2a = np.random.random(size = num_points) * 3
feature_3a = np.random.random(size = num_points) * 3
train_target = (feature_1a - 2 * feature_2a) * feature_3a + np.random.
feature_1b = np.random.random(size = num_points) * 3
feature_2b = np.random.random(size = num_points) * 3
feature_3b = np.random.random(size = num_points) * 3
test_target = (feature_1b - 2 * feature_2b) * feature_3b + np.random.n
train_df = pd.DataFrame({"f1": feature_1a, "f2": feature_2a, "f3": fea
test_df = pd.DataFrame({"f1": feature_1b, "f2": feature_2b, "f3": feat
rf = RandomForestRegressor()
rf.fit(train_df.values, train_target)
```

Out[7]:

```
▾ RandomForestRegressor
▶ Parameters
```

In [8]:

```
rmse_lst = list()
rf = RandomForestRegressor()
for x in range(round(num_points / 20), num_points, round(num_points /
    rf.fit(train_df.values[0:x,:], train_target[0:x])
    rmse_lst.append(root_mean_squared_error(rf.predict(test_df.values)

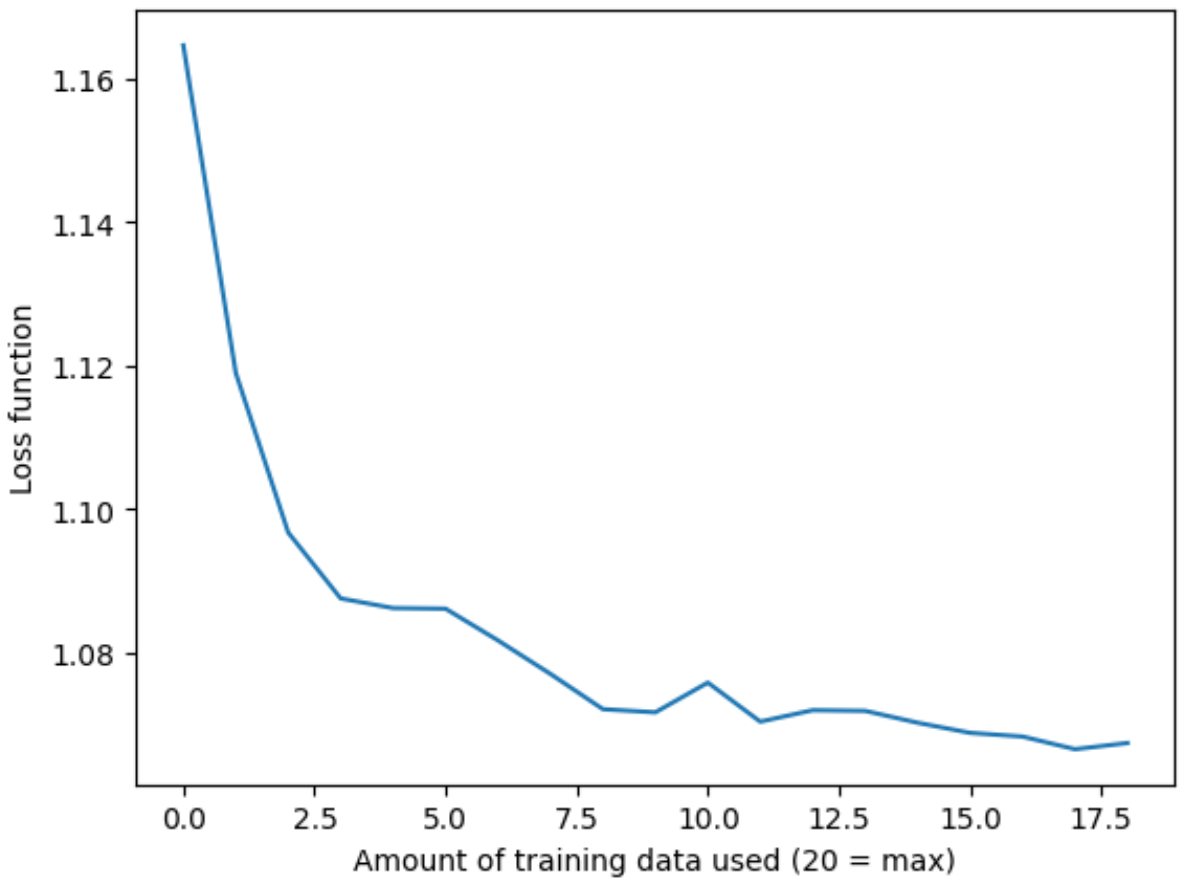
import matplotlib.pyplot as plt
```

In [9]:

```
plt.plot(rmse_lst)
plt.xlabel("Amount of training data used (20 = max)")
plt.ylabel("Loss function")
```

Out[9]:

```
Text(0, 0.5, 'Loss function')
```



In [10]:

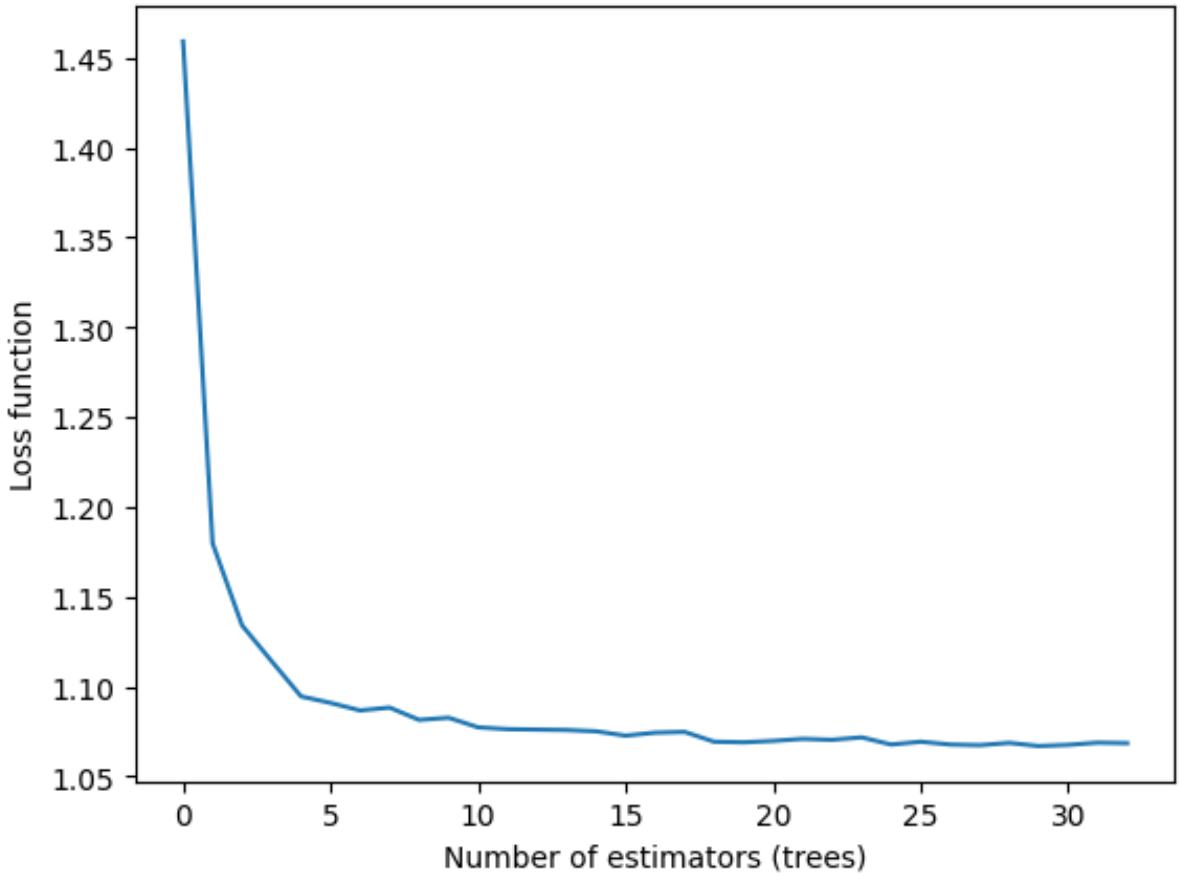
```
num_trees_lst = list()
for n_estimators in range(1, 100, 3):
    rf = RandomForestRegressor(n_estimators = n_estimators)
    rf.fit(train_df.values, train_target)
    num_trees_lst.append(root_mean_squared_error(rf.predict(test_df.va
```

In [11]:

```
plt.plot(num_trees_lst)
```

```
plt.xlabel("Number of estimators (trees)")
plt.ylabel("Loss function")
```

Out[11]: Text(0, 0.5, 'Loss function')



If I had to cut one of the plots, I would remove the Number of estimators plot. We can see past around 5 estimators, the performance does not increase much. Therefore the plot is not very informative beyond understanding that we do not need many estimators to minimize the loss.

3. Working on your datasets

This week, you will do the same types of exercises as last week, but you should use your chosen datasets that someone in your class found last semester. (They likely will not be the particular datasets that you found yourself.)

Here are some types of analysis you can do:

- Implement a random forest model.
- Perform cross-validation.
- Tune hyperparameters.
- Evaluate a performance metric.

If you like, you can try other types of models, too (beyond linear regression and random forest) although you will have many opportunities to do that next semester.

In [12]:
```
df = pd.read_csv('Milestone 3 EDA/ibm_hi_small_trans_cleaned.csv')
df
```

Out[12]:

	Timestamp	From Bank	Account	To Bank	Account.1	Amount Received
0	2022/09/01 00:20	10	8000EBD30	10	8000EBD30	3697.340000
1	2022/09/01 00:20	3208	8000F4580	1	8000F5340	0.010000
2	2022/09/01 00:00	3209	8000F4670	3209	8000F4670	14675.570000
3	2022/09/01 00:02	12	8000F5030	12	8000F5030	2806.970000
4	2022/09/01 00:06	10	8000F5200	10	8000F5200	36682.970000
...	...	...	...	...	...	...
5078340	2022/09/10 23:57	54219	8148A6631	256398	8148A8711	0.154978
5078341	2022/09/10 23:35	15	8148A8671	256398	8148A8711	0.108128
5078342	2022/09/10 23:52	154365	8148A6771	256398	8148A8711	0.004988
5078343	2022/09/10 23:46	256398	8148A6311	256398	8148A8711	0.038417
5078344	2022/09/10 23:37	154518	8148A6091	256398	8148A8711	0.281983

5078345 rows × 109 columns

In [13]:
```
df.columns
```

Out[13]:
```
Index(['Timestamp', 'From Bank', 'Account', 'To Bank', 'Account.1',
       'Amount Received', 'Amount Paid', 'Is Laundering',
       'Amount_Received_USD', 'Amount_Paid_USD',
       ...
       'Payment Currency_Yuan', 'Payment Format_ACH', 'Payment Format
_Bitcoin',
       'Payment Format_Cash', 'Payment Format_Cheque',
       'Payment Format_Credit Card', 'Payment Format_Reinvestment',
       'Payment Format_Wire', 'Account_Same', 'Bank_Same'],
      dtype='object', length=109)
```

In [14]:
```
df.drop(columns=['Timestamp', 'From Bank', 'To Bank', 'Account', 'Acco
```

In [15]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5078345 entries, 0 to 5078344
Columns: 104 entries, Amount Received to Bank_Same
dtypes: float64(101), int64(3)
memory usage: 3.9 GB
```

In [16]:
```
random_state = 42
```

In [17]:
```
fraud = df[df['Is Laundering'] == 1]
non_fraud = df[df['Is Laundering'] == 0]
len(fraud), len(non_fraud)
```

Out[17]:
```
(5177, 5073168)
```

In [18]:
```
# Undersample majority class to a 1:10 ratio
non_fraud_downsampled = non_fraud.sample(n=len(fraud) * 10, random_sta
non_fraud_downsampled
```

Out[18]:

	Amount Received	Amount Paid	Is Laundering	Amount_Received_USD	Amou
4563287	1439.79	1439.79	0	3.829841e+02	3
4216770	33.09	33.09	0	3.309000e+01	3
2992877	263.34	263.34	0	3.807896e+01	3
1499016	17154983.76	17154983.76	0	1.747064e+07	1
510282	8499.22	8499.22	0	2.501320e+03	2
...	...	...	...	...	
3992741	5722.87	5722.87	0	5.722870e+03	5
44465	82208.92	82208.92	0	8.220892e+04	8
1173046	6837.09	6837.09	0	5.196872e+03	5
4125810	3261.27	3261.27	0	3.763506e+03	3
3821365	20287.92	20287.92	0	2.933633e+03	2

51770 rows × 104 columns

In [19]:

```
# Combine and shuffle
df_reduced = pd.concat([fraud, non_fraud_downsampled]).sample(frac=1).
df_reduced
```

Out[19]:

	Amount Received	Amount Paid	Is Laundering	Amount_Received_USD	Amount_Paid
0	142.21	142.21	0	141.427845	141.4
1	8.12	8.12	0	8.269408	8.26
2	234.56	234.56	0	233.269920	233.26
3	19640.00	19640.00	0	20001.376000	20001.3
4	8.38	8.38	0	8.380000	8.38
...	...	...	...	...	
56942	78922.59	78922.59	0	3906.668205	3906.66
56943	89535.13	89535.13	0	635.699423	635.69
56944	207870.51	207870.51	0	2598.381375	2598.3
56945	29.48	29.48	0	29.317860	29.3
56946	208.99	208.99	0	30.219954	30.2

56947 rows × 104 columns

In [20]:
```
X = df_reduced.drop(columns='Is Laundering')
y = df_reduced['Is Laundering']
```

In [21]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X,
                                                    y,
                                                    test_size=0.2,
                                                    stratify=y,  # Mai
                                                    random_state=rando
                                                    )
```

In [47]:
```
scaler = StandardScaler()
```

In [48]:
```
X_train_scaled = scaler.fit_transform(X_train)
X_train_scaled
```

Out[48]:
```
array([[-0.01368816, -0.01278547, -0.01008913, ..., -0.18157169,
        -0.34846905, -0.38402302],
       [-0.01368897, -0.01278628, -0.01009458, ..., -0.18157169,
        -0.34846905, -0.38402302],
       [-0.01368909, -0.0127864 , -0.01009542, ..., -0.18157169,
        -0.34846905, -0.38402302],
       ...,
       [-0.01368367, -0.01278097, -0.01005863, ..., -0.18157169,
        -0.34846905, -0.38402302],
       [-0.01253343, -0.01162778, -0.01003977, ..., -0.18157169,
        -0.34846905, -0.38402302],
       [-0.01368867, -0.01278597, -0.01009256, ..., -0.18157169,
        -0.34846905, -0.38402302]], shape=(45557, 103))
```

In [49]:
```
X_test_scaled = scaler.transform(X_test)
```

In [25]:
```
# cv = RepeatedKFold(n_splits=5, n_repeats=5, random_state=random_stat
from sklearn.model_selection import StratifiedKFold


cv = StratifiedKFold(n_splits=5)
```

RF with scaled data & recall as param tuning metric

In [ ]:
```
# Parameter tuning
import optuna
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score


def objective(trial):
    n_estimators = trial.suggest_int("n_estimators", 10, 200)
    max_depth = trial.suggest_int("max_depth", 2, 32, log=True)
    min_samples_split = trial.suggest_int("min_samples_split", 2, 10)

    model = RandomForestClassifier(
        class_weight='balanced',
        n_estimators=n_estimators,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        random_state=random_state
    )

    scores = cross_val_score(
        model,
        X=X_train_scaled,
        y=y_train,
        cv=cv,
        # scoring="f1",
        scoring='recall',
        n_jobs=-1,
```

```
    )
    # f1 = scores.mean()
    recall = scores.mean()

    # return f1
    return recall


study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=100)

top_5_df = study.trials_dataframe().sort_values("value", ascending=Fal
display(top_5_df)
```

```
c:\Users\caleb\anaconda3\envs\iris-fisher\lib\site-packages\tqdm\auto.p
y:21: TqdmWarning: IProgress not found. Please update jupyter and ipywi
dgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.htm
l
  from .autonotebook import tqdm as notebook_tqdm
[I 2026-04-14 14:20:48,392] A new study created in memory with name: no
-name-65efc86f-277b-40e7-beb9-528fedfb91a3
[I 2026-04-14 14:20:54,968] Trial 0 finished with value: 0.708352709451
4664 and parameters: {'n_estimators': 144, 'max_depth': 28, 'min_sample
s_split': 4}. Best is trial 0 with value: 0.7083527094514664.
[I 2026-04-14 14:21:00,310] Trial 1 finished with value: 0.770645618083
6 and parameters: {'n_estimators': 148, 'max_depth': 25, 'min_samples_s
plit': 7}. Best is trial 1 with value: 0.7706456180836.
[I 2026-04-14 14:21:02,198] Trial 2 finished with value: 0.806376636772
0844 and parameters: {'n_estimators': 116, 'max_depth': 2, 'min_samples
_split': 7}. Best is trial 2 with value: 0.8063766367720844.
[I 2026-04-14 14:21:05,192] Trial 3 finished with value: 0.826893760598
591 and parameters: {'n_estimators': 171, 'max_depth': 4, 'min_samples_
split': 5}. Best is trial 3 with value: 0.826893760598591.
[I 2026-04-14 14:21:06,681] Trial 4 finished with value: 0.831964767515
7194 and parameters: {'n_estimators': 97, 'max_depth': 5, 'min_samples_
split': 6}. Best is trial 4 with value: 0.8319647675157194.
[I 2026-04-14 14:21:07,977] Trial 5 finished with value: 0.832204273818
0567 and parameters: {'n_estimators': 74, 'max_depth': 6, 'min_samples_
split': 8}. Best is trial 5 with value: 0.8322042738180567.
[I 2026-04-14 14:21:10,086] Trial 6 finished with value: 0.788753401747
0557 and parameters: {'n_estimators': 68, 'max_depth': 23, 'min_samples
_split': 8}. Best is trial 5 with value: 0.8322042738180567.
[I 2026-04-14 14:21:12,063] Trial 7 finished with value: 0.809032184752
0148 and parameters: {'n_estimators': 191, 'max_depth': 3, 'min_samples
_split': 5}. Best is trial 5 with value: 0.8322042738180567.
[I 2026-04-14 14:21:16,910] Trial 8 finished with value: 0.817483085960
0357 and parameters: {'n_estimators': 191, 'max_depth': 17, 'min_sample
s_split': 8}. Best is trial 5 with value: 0.8322042738180567.
[I 2026-04-14 14:21:18,061] Trial 9 finished with value: 0.805168907303
4854 and parameters: {'n_estimators': 125, 'max_depth': 2, 'min_samples
_split': 2}. Best is trial 5 with value: 0.8322042738180567.
[I 2026-04-14 14:21:18,729] Trial 10 finished with value: 0.83896785021
24088 and parameters: {'n_estimators': 25, 'max_depth': 10, 'min_sample
```

```
s_split': 10}. Best is trial 10 with value: 0.8389678502124088.
[I 2026-04-14 14:21:19,210] Trial 11 finished with value: 0.82448121536
33678 and parameters: {'n_estimators': 10, 'max_depth': 9, 'min_samples
_split': 10}. Best is trial 10 with value: 0.8389678502124088.
[I 2026-04-14 14:21:20,002] Trial 12 finished with value: 0.84476378618
08943 and parameters: {'n_estimators': 31, 'max_depth': 10, 'min_sample
s_split': 10}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:20,453] Trial 13 finished with value: 0.83607017359
83638 and parameters: {'n_estimators': 13, 'max_depth': 11, 'min_sample
s_split': 10}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:21,749] Trial 14 finished with value: 0.83824612623
32243 and parameters: {'n_estimators': 50, 'max_depth': 13, 'min_sample
s_split': 10}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:22,608] Trial 15 finished with value: 0.84041654283
43328 and parameters: {'n_estimators': 39, 'max_depth': 8, 'min_samples
_split': 9}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:23,497] Trial 16 finished with value: 0.83389596918
46879 and parameters: {'n_estimators': 43, 'max_depth': 7, 'min_samples
_split': 9}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:25,784] Trial 17 finished with value: 0.82593253031
7069 and parameters: {'n_estimators': 87, 'max_depth': 16, 'min_samples
_split': 9}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:26,705] Trial 18 finished with value: 0.84017528831
08105 and parameters: {'n_estimators': 44, 'max_depth': 8, 'min_samples
_split': 9}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:27,328] Trial 19 finished with value: 0.80275490521
72749 and parameters: {'n_estimators': 36, 'max_depth': 4, 'min_samples
_split': 7}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:29,138] Trial 20 finished with value: 0.82351823686
06611 and parameters: {'n_estimators': 69, 'max_depth': 15, 'min_sample
s_split': 3}. Best is trial 12 with value: 0.8447637861808943.
[I 2026-04-14 14:21:30,275] Trial 21 finished with value: 0.84741962553
10222 and parameters: {'n_estimators': 56, 'max_depth': 8, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:31,353] Trial 22 finished with value: 0.82520497893
39347 and parameters: {'n_estimators': 60, 'max_depth': 6, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:31,929] Trial 23 finished with value: 0.83728081676
89374 and parameters: {'n_estimators': 20, 'max_depth': 11, 'min_sample
s_split': 8}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:32,661] Trial 24 finished with value: 0.84500445796
40215 and parameters: {'n_estimators': 31, 'max_depth': 8, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:33,613] Trial 25 finished with value: 0.80492677866
93706 and parameters: {'n_estimators': 28, 'max_depth': 20, 'min_sample
s_split': 10}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:34,501] Trial 26 finished with value: 0.80734194623
63711 and parameters: {'n_estimators': 54, 'max_depth': 5, 'min_samples
_split': 8}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:36,417] Trial 27 finished with value: 0.83631550730
4651 and parameters: {'n_estimators': 79, 'max_depth': 13, 'min_samples
_split': 6}. Best is trial 21 with value: 0.8474196255310222.
```

```
[I 2026-04-14 14:21:37,492] Trial 28 finished with value: 0.80565170772
07275 and parameters: {'n_estimators': 92, 'max_depth': 3, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:38,121] Trial 29 finished with value: 0.81531033839
73473 and parameters: {'n_estimators': 30, 'max_depth': 6, 'min_samples
_split': 7}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:39,996] Trial 30 finished with value: 0.76436833854
88599 and parameters: {'n_estimators': 59, 'max_depth': 31, 'min_sample
s_split': 10}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:40,916] Trial 31 finished with value: 0.83800108389
71346 and parameters: {'n_estimators': 42, 'max_depth': 8, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:41,662] Trial 32 finished with value: 0.84645431606
67354 and parameters: {'n_estimators': 29, 'max_depth': 9, 'min_samples
_split': 9}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:42,383] Trial 33 finished with value: 0.83921056158
69185 and parameters: {'n_estimators': 20, 'max_depth': 12, 'min_sample
s_split': 7}. Best is trial 21 with value: 0.8474196255310222.
[I 2026-04-14 14:21:45,758] Trial 34 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 139, 'max_depth': 9, 'min_sample
s_split': 8}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:48,785] Trial 35 finished with value: 0.84790213457
80667 and parameters: {'n_estimators': 143, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:51,105] Trial 36 finished with value: 0.83679422853
91281 and parameters: {'n_estimators': 150, 'max_depth': 5, 'min_sample
s_split': 8}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:52,788] Trial 37 finished with value: 0.81458191290
36207 and parameters: {'n_estimators': 135, 'max_depth': 4, 'min_sample
s_split': 5}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:54,486] Trial 38 finished with value: 0.83969277926
37658 and parameters: {'n_estimators': 107, 'max_depth': 6, 'min_sample
s_split': 6}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:57,157] Trial 39 finished with value: 0.84910957267
64684 and parameters: {'n_estimators': 165, 'max_depth': 7, 'min_sample
s_split': 7}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:21:58,896] Trial 40 finished with value: 0.80710098308
30465 and parameters: {'n_estimators': 163, 'max_depth': 3, 'min_sample
s_split': 7}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:02,027] Trial 41 finished with value: 0.84838522636
55065 and parameters: {'n_estimators': 169, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:04,876] Trial 42 finished with value: 0.84790271731
84617 and parameters: {'n_estimators': 175, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:07,639] Trial 43 finished with value: 0.84886831815
29459 and parameters: {'n_estimators': 171, 'max_depth': 7, 'min_sample
s_split': 7}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:10,072] Trial 44 finished with value: 0.83848446705
47718 and parameters: {'n_estimators': 178, 'max_depth': 5, 'min_sample
s_split': 6}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:13,148] Trial 45 finished with value: 0.84379789397
```

```
62125 and parameters: {'n_estimators': 200, 'max_depth': 6, 'min_sample
s_split': 7}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:16,427] Trial 46 finished with value: 0.84910986404
6666 and parameters: {'n_estimators': 163, 'max_depth': 10, 'min_sample
s_split': 6}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:19,618] Trial 47 finished with value: 0.84838522636
55065 and parameters: {'n_estimators': 161, 'max_depth': 10, 'min_sampl
es_split': 5}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:24,028] Trial 48 finished with value: 0.82955513598
2471 and parameters: {'n_estimators': 185, 'max_depth': 14, 'min_sample
s_split': 4}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:27,420] Trial 49 finished with value: 0.84790271731
84617 and parameters: {'n_estimators': 159, 'max_depth': 11, 'min_sampl
es_split': 6}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:29,928] Trial 50 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 130, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 34 with value: 0.8508001025623095.
[I 2026-04-14 14:22:32,368] Trial 51 finished with value: 0.85128319434
97492 and parameters: {'n_estimators': 124, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 51 with value: 0.8512831943497492.
[I 2026-04-14 14:22:34,773] Trial 52 finished with value: 0.85128319434
97492 and parameters: {'n_estimators': 124, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 51 with value: 0.8512831943497492.
[I 2026-04-14 14:22:37,142] Trial 53 finished with value: 0.85200666655
01187 and parameters: {'n_estimators': 122, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:39,526] Trial 54 finished with value: 0.85128202886
89593 and parameters: {'n_estimators': 123, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:41,884] Trial 55 finished with value: 0.85152386613
28766 and parameters: {'n_estimators': 121, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:44,878] Trial 56 finished with value: 0.81024020559
08113 and parameters: {'n_estimators': 109, 'max_depth': 18, 'min_sampl
es_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:47,516] Trial 57 finished with value: 0.84210911231
15563 and parameters: {'n_estimators': 119, 'max_depth': 12, 'min_sampl
es_split': 5}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:49,818] Trial 58 finished with value: 0.85128202886
89593 and parameters: {'n_estimators': 119, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:52,442] Trial 59 finished with value: 0.84669586196
04552 and parameters: {'n_estimators': 123, 'max_depth': 11, 'min_sampl
es_split': 5}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:54,886] Trial 60 finished with value: 0.83052073681
69555 and parameters: {'n_estimators': 100, 'max_depth': 14, 'min_sampl
es_split': 4}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:57,091] Trial 61 finished with value: 0.85007546488
11502 and parameters: {'n_estimators': 114, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:22:59,663] Trial 62 finished with value: 0.84959208172
3513 and parameters: {'n_estimators': 137, 'max_depth': 9, 'min_samples
```

```
_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:02,316] Trial 63 finished with value: 0.84838493499
5309 and parameters: {'n_estimators': 128, 'max_depth': 10, 'min_sample
s_split': 5}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:04,953] Trial 64 finished with value: 0.84935111857
01882 and parameters: {'n_estimators': 152, 'max_depth': 8, 'min_sample
s_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:07,478] Trial 65 finished with value: 0.84307616999
7028 and parameters: {'n_estimators': 115, 'max_depth': 12, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:10,504] Trial 66 finished with value: 0.84959208172
3513 and parameters: {'n_estimators': 137, 'max_depth': 9, 'min_samples
_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:12,836] Trial 67 finished with value: 0.85007546488
11502 and parameters: {'n_estimators': 99, 'max_depth': 8, 'min_samples
_split': 5}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:15,744] Trial 68 finished with value: 0.84911015541
68633 and parameters: {'n_estimators': 122, 'max_depth': 10, 'min_sampl
es_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:19,677] Trial 69 finished with value: 0.83848796349
71417 and parameters: {'n_estimators': 144, 'max_depth': 13, 'min_sampl
es_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:22,175] Trial 70 finished with value: 0.83896785021
24089 and parameters: {'n_estimators': 132, 'max_depth': 6, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:25,485] Trial 71 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 131, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:27,923] Trial 72 finished with value: 0.84886802678
27485 and parameters: {'n_estimators': 110, 'max_depth': 8, 'min_sample
s_split': 8}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:31,004] Trial 73 finished with value: 0.85031701077
487 and parameters: {'n_estimators': 126, 'max_depth': 9, 'min_samples_
split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:34,817] Trial 74 finished with value: 0.84669557059
02578 and parameters: {'n_estimators': 140, 'max_depth': 11, 'min_sampl
es_split': 5}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:36,111] Trial 75 finished with value: 0.80516890730
34854 and parameters: {'n_estimators': 118, 'max_depth': 2, 'min_sample
s_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:38,598] Trial 76 finished with value: 0.84935082719
99908 and parameters: {'n_estimators': 104, 'max_depth': 8, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:42,519] Trial 77 finished with value: 0.84766058868
43471 and parameters: {'n_estimators': 155, 'max_depth': 10, 'min_sampl
es_split': 2}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:45,049] Trial 78 finished with value: 0.84645577291
7723 and parameters: {'n_estimators': 93, 'max_depth': 12, 'min_samples
_split': 8}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:47,642] Trial 79 finished with value: 0.84814338910
15892 and parameters: {'n_estimators': 129, 'max_depth': 7, 'min_sample
s_split': 7}. Best is trial 53 with value: 0.8520066665501187.
```

```
[I 2026-04-14 14:23:52,012] Trial 80 finished with value: 0.82714084252
60631 and parameters: {'n_estimators': 146, 'max_depth': 15, 'min_sampl
es_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:55,058] Trial 81 finished with value: 0.85007546488
11502 and parameters: {'n_estimators': 133, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:23:57,841] Trial 82 finished with value: 0.85152386613
28766 and parameters: {'n_estimators': 121, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:00,389] Trial 83 finished with value: 0.85031701077
487 and parameters: {'n_estimators': 113, 'max_depth': 8, 'min_samples_
split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:03,247] Trial 84 finished with value: 0.84935140994
03857 and parameters: {'n_estimators': 122, 'max_depth': 10, 'min_sampl
es_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:06,037] Trial 85 finished with value: 0.84886773541
2551 and parameters: {'n_estimators': 140, 'max_depth': 7, 'min_samples
_split': 7}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:08,837] Trial 86 finished with value: 0.84790300868
86592 and parameters: {'n_estimators': 105, 'max_depth': 11, 'min_sampl
es_split': 8}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:11,079] Trial 87 finished with value: 0.84017470557
04154 and parameters: {'n_estimators': 119, 'max_depth': 6, 'min_sample
s_split': 6}. Best is trial 53 with value: 0.8520066665501187.
[I 2026-04-14 14:24:13,947] Trial 88 finished with value: 0.85249004970
77559 and parameters: {'n_estimators': 124, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:18,147] Trial 89 finished with value: 0.73949960082
28294 and parameters: {'n_estimators': 112, 'max_depth': 25, 'min_sampl
es_split': 4}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:20,347] Trial 90 finished with value: 0.84669557059
02578 and parameters: {'n_estimators': 82, 'max_depth': 11, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:23,184] Trial 91 finished with value: 0.85224850381
40359 and parameters: {'n_estimators': 126, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:25,952] Trial 92 finished with value: 0.85031701077
487 and parameters: {'n_estimators': 126, 'max_depth': 8, 'min_samples_
split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:28,674] Trial 93 finished with value: 0.85176570339
67938 and parameters: {'n_estimators': 118, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:31,645] Trial 94 finished with value: 0.85031730214
50674 and parameters: {'n_estimators': 120, 'max_depth': 10, 'min_sampl
es_split': 4}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:34,170] Trial 95 finished with value: 0.84935111857
01882 and parameters: {'n_estimators': 125, 'max_depth': 7, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:36,727] Trial 96 finished with value: 0.85200754066
0711 and parameters: {'n_estimators': 103, 'max_depth': 9, 'min_samples
_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:39,584] Trial 97 finished with value: 0.84283345862
```

```
25181 and parameters: {'n_estimators': 101, 'max_depth': 12, 'min_sampl
es_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:41,903] Trial 98 finished with value: 0.85007575625
13476 and parameters: {'n_estimators': 108, 'max_depth': 8, 'min_sample
s_split': 5}. Best is trial 88 with value: 0.8524900497077559.
[I 2026-04-14 14:24:44,238] Trial 99 finished with value: 0.84838551773
57039 and parameters: {'n_estimators': 94, 'max_depth': 10, 'min_sample
s_split': 4}. Best is trial 88 with value: 0.8524900497077559.
```

	number	value	datetime_start	datetime_complete	duration	pa
88	88	0.852490	2026-04-14 14:24:11.079564	2026-04-14 14:24:13.947634	0 days 00:00:02.868070	
91	91	0.852249	2026-04-14 14:24:20.348595	2026-04-14 14:24:23.184013	0 days 00:00:02.835418	
96	96	0.852008	2026-04-14 14:24:34.171165	2026-04-14 14:24:36.727563	0 days 00:00:02.556398	
53	53	0.852007	2026-04-14 14:22:34.775041	2026-04-14 14:22:37.142433	0 days 00:00:02.367392	
93	93	0.851766	2026-04-14 14:24:25.953199	2026-04-14 14:24:28.674108	0 days 00:00:02.720909	

In [43]:
```
# Train using optimal parameters
model = RandomForestClassifier(
    class_weight='balanced',
    n_estimators=top_5_df.head(1)['params_n_estimators'].item(),
    max_depth=top_5_df.head(1)['params_max_depth'].item(),
    min_samples_split=top_5_df.head(1)['params_min_samples_split'].ite
    random_state=random_state
)
model.fit(X_train_scaled, y_train)
```

Out[43]:

```
▾ RandomForestClassifier
▶ Parameters
```

In [50]:
```
y_pred = model.predict(X_test_scaled)
```

In [51]:
```
print(classification_report(y_true=y_test, y_pred=y_pred))
```

```
              precision    recall  f1-score   support

           0       0.98      0.93      0.96     10355
           1       0.55      0.85      0.67      1035

    accuracy                           0.92     11390
   macro avg       0.77      0.89      0.81     11390
weighted avg       0.94      0.92      0.93     11390
```

RF With unscaled Data and recall as param tuning metric

In [58]:
```
def objective(trial):
    n_estimators = trial.suggest_int("n_estimators", 10, 200)
    max_depth = trial.suggest_int("max_depth", 2, 32, log=True)
    min_samples_split = trial.suggest_int("min_samples_split", 2, 10)

    model = RandomForestClassifier(
        class_weight='balanced',
        n_estimators=n_estimators,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        random_state=random_state
    )

    scores = cross_val_score(
        model,
        X=X_train,
        y=y_train,
        cv=cv,
        scoring='recall',
        n_jobs=-1,
    )
    recall = scores.mean()

    return recall


study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=100)

top_5_unscaled_df = study.trials_dataframe().sort_values("value", asce
display(top_5_unscaled_df)
```

```
[I 2026-04-14 14:51:01,029] A new study created in memory with name: no
-name-81346362-09b7-423e-a1de-c904b184b1e7
[I 2026-04-14 14:51:01,605] Trial 0 finished with value: 0.816273025529
8568 and parameters: {'n_estimators': 75, 'max_depth': 2, 'min_samples_
split': 6}. Best is trial 0 with value: 0.8162730255298568.
[I 2026-04-14 14:51:02,244] Trial 1 finished with value: 0.804203306469
0012 and parameters: {'n_estimators': 61, 'max_depth': 3, 'min_samples_
split': 9}. Best is trial 0 with value: 0.8162730255298568.
```

```
[I 2026-04-14 14:51:04,307] Trial 2 finished with value: 0.823276690966
9411 and parameters: {'n_estimators': 71, 'max_depth': 17, 'min_samples
_split': 10}. Best is trial 2 with value: 0.8232766909669411.
[I 2026-04-14 14:51:05,011] Trial 3 finished with value: 0.804686106886
2432 and parameters: {'n_estimators': 49, 'max_depth': 5, 'min_samples_
split': 5}. Best is trial 2 with value: 0.8232766909669411.
[I 2026-04-14 14:51:05,320] Trial 4 finished with value: 0.839936073378
6704 and parameters: {'n_estimators': 14, 'max_depth': 2, 'min_samples_
split': 7}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:06,147] Trial 5 finished with value: 0.809514985169
257 and parameters: {'n_estimators': 75, 'max_depth': 4, 'min_samples_s
plit': 4}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:06,724] Trial 6 finished with value: 0.816997371840
8187 and parameters: {'n_estimators': 70, 'max_depth': 2, 'min_samples_
split': 10}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:07,239] Trial 7 finished with value: 0.808549384334
7727 and parameters: {'n_estimators': 59, 'max_depth': 2, 'min_samples_
split': 9}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:08,004] Trial 8 finished with value: 0.835586499070
5291 and parameters: {'n_estimators': 44, 'max_depth': 7, 'min_samples_
split': 2}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:08,785] Trial 9 finished with value: 0.816277396082
819 and parameters: {'n_estimators': 22, 'max_depth': 17, 'min_samples_
split': 7}. Best is trial 4 with value: 0.8399360733786704.
[I 2026-04-14 14:51:11,434] Trial 10 finished with value: 0.85128232023
91566 and parameters: {'n_estimators': 157, 'max_depth': 9, 'min_sample
s_split': 7}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:14,570] Trial 11 finished with value: 0.84814338910
15892 and parameters: {'n_estimators': 176, 'max_depth': 10, 'min_sampl
es_split': 7}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:18,043] Trial 12 finished with value: 0.84597064153
89009 and parameters: {'n_estimators': 176, 'max_depth': 11, 'min_sampl
es_split': 7}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:20,793] Trial 13 finished with value: 0.85079981119
2112 and parameters: {'n_estimators': 167, 'max_depth': 9, 'min_samples
_split': 8}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:24,885] Trial 14 finished with value: 0.78826972721
92212 and parameters: {'n_estimators': 138, 'max_depth': 23, 'min_sampl
es_split': 8}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:27,039] Trial 15 finished with value: 0.85055855666
85896 and parameters: {'n_estimators': 139, 'max_depth': 8, 'min_sample
s_split': 4}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:30,360] Trial 16 finished with value: 0.83269290163
92487 and parameters: {'n_estimators': 144, 'max_depth': 14, 'min_sampl
es_split': 8}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:32,461] Trial 17 finished with value: 0.83751915759
0485 and parameters: {'n_estimators': 198, 'max_depth': 5, 'min_samples
_split': 5}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:35,859] Trial 18 finished with value: 0.75929733163
17313 and parameters: {'n_estimators': 109, 'max_depth': 28, 'min_sampl
es_split': 8}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:37,892] Trial 19 finished with value: 0.84790242594
```

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82644 and parameters: {'n_estimators': 108, 'max_depth': 10, 'min_sampl
es_split': 6}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:39,905] Trial 20 finished with value: 0.84428011165
30596 and parameters: {'n_estimators': 162, 'max_depth': 6, 'min_sample
s_split': 9}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:41,999] Trial 21 finished with value: 0.84862618951
88314 and parameters: {'n_estimators': 135, 'max_depth': 8, 'min_sample
s_split': 3}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:44,372] Trial 22 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 158, 'max_depth': 8, 'min_sample
s_split': 4}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:47,562] Trial 23 finished with value: 0.84235036683
50787 and parameters: {'n_estimators': 156, 'max_depth': 12, 'min_sampl
es_split': 5}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:49,367] Trial 24 finished with value: 0.83293095109
05987 and parameters: {'n_estimators': 199, 'max_depth': 4, 'min_sample
s_split': 3}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:51,362] Trial 25 finished with value: 0.84935111857
01881 and parameters: {'n_estimators': 124, 'max_depth': 8, 'min_sample
s_split': 6}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:55,323] Trial 26 finished with value: 0.82979668187
61908 and parameters: {'n_estimators': 173, 'max_depth': 14, 'min_sampl
es_split': 4}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:51:58,100] Trial 27 finished with value: 0.76026526342
77955 and parameters: {'n_estimators': 92, 'max_depth': 21, 'min_sample
s_split': 2}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:52:00,132] Trial 28 finished with value: 0.84452224028
71744 and parameters: {'n_estimators': 163, 'max_depth': 6, 'min_sample
s_split': 8}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:52:01,552] Trial 29 finished with value: 0.80903218475
20148 and parameters: {'n_estimators': 185, 'max_depth': 3, 'min_sample
s_split': 6}. Best is trial 10 with value: 0.8512823202391566.
[I 2026-04-14 14:52:03,643] Trial 30 finished with value: 0.85128232023
91567 and parameters: {'n_estimators': 120, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 30 with value: 0.8512823202391567.
[I 2026-04-14 14:52:05,796] Trial 31 finished with value: 0.85152357476
26789 and parameters: {'n_estimators': 122, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:08,352] Trial 32 finished with value: 0.84307558725
6633 and parameters: {'n_estimators': 120, 'max_depth': 12, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:10,227] Trial 33 finished with value: 0.84210794683
07664 and parameters: {'n_estimators': 149, 'max_depth': 6, 'min_sample
s_split': 3}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:12,682] Trial 34 finished with value: 0.81796617774
74753 and parameters: {'n_estimators': 90, 'max_depth': 16, 'min_sample
s_split': 4}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:14,456] Trial 35 finished with value: 0.84910928130
62708 and parameters: {'n_estimators': 125, 'max_depth': 7, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:15,613] Trial 36 finished with value: 0.82761868964
99478 and parameters: {'n_estimators': 98, 'max_depth': 5, 'min_samples
```

```
_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:18,140] Trial 37 finished with value: 0.85104106571
56344 and parameters: {'n_estimators': 151, 'max_depth': 9, 'min_sample
s_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:19,267] Trial 38 finished with value: 0.81216936766
83975 and parameters: {'n_estimators': 117, 'max_depth': 4, 'min_sample
s_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:22,512] Trial 39 finished with value: 0.83752265403
28548 and parameters: {'n_estimators': 149, 'max_depth': 13, 'min_sampl
es_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:24,972] Trial 40 finished with value: 0.84790271731
84617 and parameters: {'n_estimators': 129, 'max_depth': 10, 'min_sampl
es_split': 7}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:27,560] Trial 41 finished with value: 0.84983304487
68379 and parameters: {'n_estimators': 155, 'max_depth': 9, 'min_sample
s_split': 4}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:30,273] Trial 42 finished with value: 0.84959208172
35131 and parameters: {'n_estimators': 189, 'max_depth': 7, 'min_sample
s_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:32,646] Trial 43 finished with value: 0.84886860952
31437 and parameters: {'n_estimators': 152, 'max_depth': 8, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:35,820] Trial 44 finished with value: 0.84524746070
87289 and parameters: {'n_estimators': 134, 'max_depth': 11, 'min_sampl
es_split': 7}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:38,185] Trial 45 finished with value: 0.85128232023
91567 and parameters: {'n_estimators': 115, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:40,384] Trial 46 finished with value: 0.82327756507
75336 and parameters: {'n_estimators': 82, 'max_depth': 16, 'min_sample
s_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:42,362] Trial 47 finished with value: 0.85055768255
79974 and parameters: {'n_estimators': 112, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:43,767] Trial 48 finished with value: 0.84114147188
56897 and parameters: {'n_estimators': 102, 'max_depth': 6, 'min_sample
s_split': 7}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:46,451] Trial 49 finished with value: 0.84935140994
03857 and parameters: {'n_estimators': 141, 'max_depth': 10, 'min_sampl
es_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:47,965] Trial 50 finished with value: 0.84404060535
07223 and parameters: {'n_estimators': 64, 'max_depth': 12, 'min_sample
s_split': 10}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:50,294] Trial 51 finished with value: 0.84959179035
33154 and parameters: {'n_estimators': 169, 'max_depth': 7, 'min_sample
s_split': 4}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:52,514] Trial 52 finished with value: 0.84983304487
68379 and parameters: {'n_estimators': 130, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:54,384] Trial 53 finished with value: 0.84959237309
37104 and parameters: {'n_estimators': 117, 'max_depth': 8, 'min_sample
s_split': 4}. Best is trial 31 with value: 0.8515235747626789.
```

```
[I 2026-04-14 14:52:57,868] Trial 54 finished with value: 0.84548784112
16586 and parameters: {'n_estimators': 182, 'max_depth': 11, 'min_sampl
es_split': 3}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:52:59,739] Trial 55 finished with value: 0.84186610956
6849 and parameters: {'n_estimators': 145, 'max_depth': 6, 'min_samples
_split': 7}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:02,436] Trial 56 finished with value: 0.84983333624
70353 and parameters: {'n_estimators': 161, 'max_depth': 9, 'min_sample
s_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:05,355] Trial 57 finished with value: 0.80227210480
00327 and parameters: {'n_estimators': 103, 'max_depth': 18, 'min_sampl
es_split': 4}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:08,065] Trial 58 finished with value: 0.83221068396
24016 and parameters: {'n_estimators': 113, 'max_depth': 14, 'min_sampl
es_split': 6}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:10,017] Trial 59 finished with value: 0.84886773541
2551 and parameters: {'n_estimators': 138, 'max_depth': 7, 'min_samples
_split': 5}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:10,501] Trial 60 finished with value: 0.80275519658
74723 and parameters: {'n_estimators': 26, 'max_depth': 5, 'min_samples
_split': 4}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:13,558] Trial 61 finished with value: 0.84910928130
62708 and parameters: {'n_estimators': 167, 'max_depth': 10, 'min_sampl
es_split': 8}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:16,309] Trial 62 finished with value: 0.85055797392
81947 and parameters: {'n_estimators': 178, 'max_depth': 8, 'min_sample
s_split': 9}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:19,486] Trial 63 finished with value: 0.84645373332
63404 and parameters: {'n_estimators': 159, 'max_depth': 11, 'min_sampl
es_split': 7}. Best is trial 31 with value: 0.8515235747626789.
[I 2026-04-14 14:53:22,685] Trial 64 finished with value: 0.85176570339
67938 and parameters: {'n_estimators': 193, 'max_depth': 9, 'min_sample
s_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:25,517] Trial 65 finished with value: 0.85031613666
42775 and parameters: {'n_estimators': 189, 'max_depth': 8, 'min_sample
s_split': 7}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:27,774] Trial 66 finished with value: 0.85079981119
2112 and parameters: {'n_estimators': 124, 'max_depth': 9, 'min_samples
_split': 9}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:30,450] Trial 67 finished with value: 0.84959179035
33154 and parameters: {'n_estimators': 194, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:33,567] Trial 68 finished with value: 0.84162776874
53017 and parameters: {'n_estimators': 146, 'max_depth': 12, 'min_sampl
es_split': 6}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:34,863] Trial 69 finished with value: 0.83872630431
86891 and parameters: {'n_estimators': 95, 'max_depth': 6, 'min_samples
_split': 5}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:38,012] Trial 70 finished with value: 0.83221097533
25991 and parameters: {'n_estimators': 132, 'max_depth': 14, 'min_sampl
es_split': 3}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:40,987] Trial 71 finished with value: 0.85176570339
```

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67938 and parameters: {'n_estimators': 175, 'max_depth': 9, 'min_sample
s_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:44,107] Trial 72 finished with value: 0.84935053582
9793 and parameters: {'n_estimators': 173, 'max_depth': 10, 'min_sample
s_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:47,049] Trial 73 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 178, 'max_depth': 9, 'min_sample
s_split': 9}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:50,539] Trial 74 finished with value: 0.84476436892
12893 and parameters: {'n_estimators': 184, 'max_depth': 11, 'min_sampl
es_split': 7}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:52,803] Trial 75 finished with value: 0.84935053582
97933 and parameters: {'n_estimators': 166, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:56,898] Trial 76 finished with value: 0.83703956224
54152 and parameters: {'n_estimators': 193, 'max_depth': 13, 'min_sampl
es_split': 5}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:53:58,319] Trial 77 finished with value: 0.84838522636
55065 and parameters: {'n_estimators': 84, 'max_depth': 8, 'min_samples
_split': 8}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:54:01,172] Trial 78 finished with value: 0.84790242594
82643 and parameters: {'n_estimators': 157, 'max_depth': 10, 'min_sampl
es_split': 7}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:54:02,130] Trial 79 finished with value: 0.80951498516
9257 and parameters: {'n_estimators': 152, 'max_depth': 2, 'min_samples
_split': 5}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:54:07,400] Trial 80 finished with value: 0.71390622541
56396 and parameters: {'n_estimators': 172, 'max_depth': 32, 'min_sampl
es_split': 6}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:54:10,361] Trial 81 finished with value: 0.85080010256
23095 and parameters: {'n_estimators': 178, 'max_depth': 9, 'min_sample
s_split': 9}. Best is trial 64 with value: 0.8517657033967938.
[I 2026-04-14 14:54:13,319] Trial 82 finished with value: 0.85200695792
0316 and parameters: {'n_estimators': 181, 'max_depth': 9, 'min_samples
_split': 10}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:15,026] Trial 83 finished with value: 0.84935140994
03857 and parameters: {'n_estimators': 108, 'max_depth': 8, 'min_sample
s_split': 10}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:17,680] Trial 84 finished with value: 0.84959149898
31179 and parameters: {'n_estimators': 196, 'max_depth': 7, 'min_sample
s_split': 10}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:20,992] Trial 85 finished with value: 0.84910928130
62708 and parameters: {'n_estimators': 188, 'max_depth': 10, 'min_sampl
es_split': 6}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:23,391] Trial 86 finished with value: 0.84766058868
43471 and parameters: {'n_estimators': 122, 'max_depth': 11, 'min_sampl
es_split': 9}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:26,949] Trial 87 finished with value: 0.83703985361
56127 and parameters: {'n_estimators': 165, 'max_depth': 13, 'min_sampl
es_split': 5}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:30,241] Trial 88 finished with value: 0.84935053582
97933 and parameters: {'n_estimators': 200, 'max_depth': 9, 'min_sample
```

```
s_split': 4}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:32,454] Trial 89 finished with value: 0.84983391898
74302 and parameters: {'n_estimators': 141, 'max_depth': 8, 'min_sample
s_split': 10}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:33,887] Trial 90 finished with value: 0.82858545596
5222 and parameters: {'n_estimators': 128, 'max_depth': 5, 'min_samples
_split': 6}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:37,050] Trial 91 finished with value: 0.85079951982
19145 and parameters: {'n_estimators': 180, 'max_depth': 10, 'min_sampl
es_split': 9}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:39,923] Trial 92 finished with value: 0.84983391898
74302 and parameters: {'n_estimators': 171, 'max_depth': 9, 'min_sample
s_split': 9}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:42,850] Trial 93 finished with value: 0.85176570339
67938 and parameters: {'n_estimators': 175, 'max_depth': 9, 'min_sample
s_split': 8}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:45,375] Trial 94 finished with value: 0.84983304487
68379 and parameters: {'n_estimators': 185, 'max_depth': 7, 'min_sample
s_split': 8}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:48,481] Trial 95 finished with value: 0.84524658659
81365 and parameters: {'n_estimators': 162, 'max_depth': 11, 'min_sampl
es_split': 8}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:50,808] Trial 96 finished with value: 0.84959208172
3513 and parameters: {'n_estimators': 153, 'max_depth': 8, 'min_samples
_split': 7}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:53,633] Trial 97 finished with value: 0.85176570339
67938 and parameters: {'n_estimators': 175, 'max_depth': 9, 'min_sample
s_split': 8}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:54:57,528] Trial 98 finished with value: 0.84307529588
64355 and parameters: {'n_estimators': 191, 'max_depth': 12, 'min_sampl
es_split': 8}. Best is trial 82 with value: 0.852006957920316.
[I 2026-04-14 14:55:00,618] Trial 99 finished with value: 0.84790184320
78693 and parameters: {'n_estimators': 175, 'max_depth': 10, 'min_sampl
es_split': 7}. Best is trial 82 with value: 0.852006957920316.
```

	number	value	datetime_start	datetime_complete	duration	pa
82	82	0.852007	2026-04-14 14:54:10.362024	2026-04-14 14:54:13.319674	0 days 00:00:02.957650	
93	93	0.851766	2026-04-14 14:54:39.923495	2026-04-14 14:54:42.850984	0 days 00:00:02.927489	
64	64	0.851766	2026-04-14 14:53:19.486939	2026-04-14 14:53:22.685829	0 days 00:00:03.198890	
71	71	0.851766	2026-04-14 14:53:38.013142	2026-04-14 14:53:40.987306	0 days 00:00:02.974164	
97	97	0.851766	2026-04-14 14:54:50.809066	2026-04-14 14:54:53.633662	0 days 00:00:02.824596	

In [59]:
```
# Train using optimal parameters
model = RandomForestClassifier(
```

```
    class_weight='balanced',
    n_estimators=top_5_unscaled_df.head(1)['params_n_estimators'].item
    max_depth=top_5_unscaled_df.head(1)['params_max_depth'].item(),
    min_samples_split=top_5_unscaled_df.head(1)['params_min_samples_sp
    random_state=random_state
)
model.fit(X_train, y_train)
```

Out[59]:

```
▾ RandomForestClassifier
▶ Parameters
```

In [60]:

```
y_pred = model.predict(X_test)
```

In [61]:

```
print(classification_report(y_true=y_test, y_pred=y_pred))
```

```
              precision    recall  f1-score   support

           0       0.98      0.93      0.96     10355
           1       0.55      0.84      0.67      1035

    accuracy                           0.92     11390
   macro avg       0.77      0.89      0.81     11390
weighted avg       0.94      0.92      0.93     11390
```

In [62]:

```
recall_model_report_df = classification_report(y_test, y_pred, output_
```

No scaling, f1 as eval metric

In [ ]:

```
# Parameter tuning
import optuna
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score


def objective(trial):
    n_estimators = trial.suggest_int("n_estimators", 10, 200)
    max_depth = trial.suggest_int("max_depth", 2, 32, log=True)
    min_samples_split = trial.suggest_int("min_samples_split", 2, 10)

    model = RandomForestClassifier(
        class_weight='balanced',
        n_estimators=n_estimators,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        random_state=random_state
    )

    scores = cross_val_score(
```

```
        model,
        X=X_train,
        y=y_train,
        cv=cv,
        scoring="f1",
        n_jobs=-1,
    )
    f1 = scores.mean()
    return f1


study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=100)

top_5_f1_df = study.trials_dataframe().sort_values("value", ascending=
display(top_5_f1_df)
```

```
[I 2026-04-14 14:44:46,858] A new study created in memory with name: no
-name-bd465fdb-5a7e-40a5-8419-cd9e25ede6c2
[I 2026-04-14 14:44:48,804] Trial 0 finished with value: 0.668916432371
8346 and parameters: {'n_estimators': 35, 'max_depth': 5, 'min_samples_
split': 10}. Best is trial 0 with value: 0.6689164323718346.
[I 2026-04-14 14:44:50,992] Trial 1 finished with value: 0.667854481318
131 and parameters: {'n_estimators': 115, 'max_depth': 4, 'min_samples_
split': 4}. Best is trial 0 with value: 0.6689164323718346.
[I 2026-04-14 14:44:54,599] Trial 2 finished with value: 0.690574256352
4027 and parameters: {'n_estimators': 92, 'max_depth': 18, 'min_samples
_split': 3}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:44:56,789] Trial 3 finished with value: 0.665517972995
3378 and parameters: {'n_estimators': 95, 'max_depth': 5, 'min_samples_
split': 5}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:44:59,206] Trial 4 finished with value: 0.688823970778
2965 and parameters: {'n_estimators': 81, 'max_depth': 20, 'min_samples
_split': 6}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:00,215] Trial 5 finished with value: 0.668958088869
8118 and parameters: {'n_estimators': 81, 'max_depth': 5, 'min_samples_
split': 10}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:01,429] Trial 6 finished with value: 0.678945179261
0242 and parameters: {'n_estimators': 43, 'max_depth': 12, 'min_samples
_split': 2}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:02,135] Trial 7 finished with value: 0.669331172377
9061 and parameters: {'n_estimators': 57, 'max_depth': 4, 'min_samples_
split': 6}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:03,600] Trial 8 finished with value: 0.660867102479
0465 and parameters: {'n_estimators': 97, 'max_depth': 7, 'min_samples_
split': 7}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:08,973] Trial 9 finished with value: 0.657292373253
9245 and parameters: {'n_estimators': 173, 'max_depth': 26, 'min_sample
s_split': 3}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:09,952] Trial 10 finished with value: 0.64613355384
07518 and parameters: {'n_estimators': 148, 'max_depth': 2, 'min_sample
s_split': 2}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:10,513] Trial 11 finished with value: 0.65671323080
```

```
15359 and parameters: {'n_estimators': 11, 'max_depth': 29, 'min_sample
s_split': 8}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:13,953] Trial 12 finished with value: 0.68856036819
39719 and parameters: {'n_estimators': 130, 'max_depth': 17, 'min_sampl
es_split': 5}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:15,758] Trial 13 finished with value: 0.68525060202
31589 and parameters: {'n_estimators': 69, 'max_depth': 15, 'min_sample
s_split': 8}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:19,006] Trial 14 finished with value: 0.66213132542
61243 and parameters: {'n_estimators': 189, 'max_depth': 9, 'min_sample
s_split': 4}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:23,012] Trial 15 finished with value: 0.67523656364
19118 and parameters: {'n_estimators': 132, 'max_depth': 24, 'min_sampl
es_split': 6}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:25,058] Trial 16 finished with value: 0.68808229703
87469 and parameters: {'n_estimators': 80, 'max_depth': 15, 'min_sample
s_split': 3}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:28,475] Trial 17 finished with value: 0.68349842317
46521 and parameters: {'n_estimators': 114, 'max_depth': 22, 'min_sampl
es_split': 8}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:31,222] Trial 18 finished with value: 0.66452700839
73858 and parameters: {'n_estimators': 149, 'max_depth': 10, 'min_sampl
es_split': 4}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:32,623] Trial 19 finished with value: 0.68722144400
61214 and parameters: {'n_estimators': 44, 'max_depth': 19, 'min_sample
s_split': 6}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:32,917] Trial 20 finished with value: 0.29414649526
33757 and parameters: {'n_estimators': 13, 'max_depth': 2, 'min_samples
_split': 9}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:37,225] Trial 21 finished with value: 0.64003962388
78596 and parameters: {'n_estimators': 135, 'max_depth': 32, 'min_sampl
es_split': 5}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:40,001] Trial 22 finished with value: 0.68725714451
29589 and parameters: {'n_estimators': 107, 'max_depth': 15, 'min_sampl
es_split': 5}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:43,654] Trial 23 finished with value: 0.69029433851
80034 and parameters: {'n_estimators': 129, 'max_depth': 19, 'min_sampl
es_split': 3}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:45,312] Trial 24 finished with value: 0.66989496118
28121 and parameters: {'n_estimators': 79, 'max_depth': 11, 'min_sample
s_split': 3}. Best is trial 2 with value: 0.6905742563524027.
[I 2026-04-14 14:45:49,990] Trial 25 finished with value: 0.69173569198
27584 and parameters: {'n_estimators': 164, 'max_depth': 19, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:45:52,472] Trial 26 finished with value: 0.66286413491
69481 and parameters: {'n_estimators': 161, 'max_depth': 8, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:45:56,680] Trial 27 finished with value: 0.68519175915
44909 and parameters: {'n_estimators': 193, 'max_depth': 13, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:01,559] Trial 28 finished with value: 0.68866747487
56307 and parameters: {'n_estimators': 174, 'max_depth': 20, 'min_sampl
```

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es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:05,328] Trial 29 finished with value: 0.66442610614
67609 and parameters: {'n_estimators': 123, 'max_depth': 25, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:08,547] Trial 30 finished with value: 0.68320942976
4161 and parameters: {'n_estimators': 147, 'max_depth': 13, 'min_sample
s_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:11,157] Trial 31 finished with value: 0.68871763860
88202 and parameters: {'n_estimators': 94, 'max_depth': 18, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:13,107] Trial 32 finished with value: 0.68582775333
56544 and parameters: {'n_estimators': 62, 'max_depth': 20, 'min_sample
s_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:16,470] Trial 33 finished with value: 0.63856971718
87286 and parameters: {'n_estimators': 105, 'max_depth': 28, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:18,808] Trial 34 finished with value: 0.68703146897
02013 and parameters: {'n_estimators': 87, 'max_depth': 16, 'min_sample
s_split': 7}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:22,260] Trial 35 finished with value: 0.67996676396
35387 and parameters: {'n_estimators': 119, 'max_depth': 22, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:23,649] Trial 36 finished with value: 0.65225434692
75287 and parameters: {'n_estimators': 163, 'max_depth': 3, 'min_sample
s_split': 10}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:26,041] Trial 37 finished with value: 0.60722294990
08124 and parameters: {'n_estimators': 70, 'max_depth': 32, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:27,271] Trial 38 finished with value: 0.66352486056
29753 and parameters: {'n_estimators': 91, 'max_depth': 6, 'min_samples
_split': 7}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:28,520] Trial 39 finished with value: 0.68408217720
17329 and parameters: {'n_estimators': 48, 'max_depth': 12, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:29,238] Trial 40 finished with value: 0.66600618931
81288 and parameters: {'n_estimators': 31, 'max_depth': 9, 'min_samples
_split': 6}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:31,973] Trial 41 finished with value: 0.68925275994
45065 and parameters: {'n_estimators': 100, 'max_depth': 18, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:35,206] Trial 42 finished with value: 0.67312845952
50001 and parameters: {'n_estimators': 106, 'max_depth': 23, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:38,944] Trial 43 finished with value: 0.68867857902
98859 and parameters: {'n_estimators': 141, 'max_depth': 17, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:41,258] Trial 44 finished with value: 0.68576094023
42912 and parameters: {'n_estimators': 98, 'max_depth': 14, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:44,444] Trial 45 finished with value: 0.68823392508
39001 and parameters: {'n_estimators': 113, 'max_depth': 20, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
```

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[I 2026-04-14 14:46:46,822] Trial 46 finished with value: 0.65628998109
53275 and parameters: {'n_estimators': 73, 'max_depth': 27, 'min_sample
s_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:50,608] Trial 47 finished with value: 0.67001223354
42368 and parameters: {'n_estimators': 200, 'max_depth': 11, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:54,098] Trial 48 finished with value: 0.68725014613
15146 and parameters: {'n_estimators': 127, 'max_depth': 18, 'min_sampl
es_split': 9}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:56,018] Trial 49 finished with value: 0.67838421390
23621 and parameters: {'n_estimators': 58, 'max_depth': 22, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:46:57,780] Trial 50 finished with value: 0.65583310951
38449 and parameters: {'n_estimators': 178, 'max_depth': 4, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:00,829] Trial 51 finished with value: 0.68906518929
61971 and parameters: {'n_estimators': 97, 'max_depth': 17, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:03,875] Trial 52 finished with value: 0.68930755085
98343 and parameters: {'n_estimators': 99, 'max_depth': 16, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:06,987] Trial 53 finished with value: 0.68945870453
87677 and parameters: {'n_estimators': 101, 'max_depth': 16, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:09,578] Trial 54 finished with value: 0.68611585570
78625 and parameters: {'n_estimators': 88, 'max_depth': 14, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:12,837] Trial 55 finished with value: 0.68884481077
84508 and parameters: {'n_estimators': 103, 'max_depth': 15, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:15,967] Trial 56 finished with value: 0.67878508491
98002 and parameters: {'n_estimators': 116, 'max_depth': 12, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:22,326] Trial 57 finished with value: 0.65536886146
71558 and parameters: {'n_estimators': 158, 'max_depth': 25, 'min_sampl
es_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:24,373] Trial 58 finished with value: 0.66408249234
68774 and parameters: {'n_estimators': 83, 'max_depth': 10, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:27,787] Trial 59 finished with value: 0.68943913832
0691 and parameters: {'n_estimators': 110, 'max_depth': 16, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:30,028] Trial 60 finished with value: 0.66505033049
67457 and parameters: {'n_estimators': 127, 'max_depth': 6, 'min_sample
s_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:33,475] Trial 61 finished with value: 0.68942955736
52645 and parameters: {'n_estimators': 112, 'max_depth': 16, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:37,082] Trial 62 finished with value: 0.68942955736
52645 and parameters: {'n_estimators': 112, 'max_depth': 16, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:41,046] Trial 63 finished with value: 0.68504081840
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87634 and parameters: {'n_estimators': 136, 'max_depth': 13, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:44,308] Trial 64 finished with value: 0.68863821966
1238 and parameters: {'n_estimators': 112, 'max_depth': 20, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:47,127] Trial 65 finished with value: 0.68923043623
23524 and parameters: {'n_estimators': 109, 'max_depth': 16, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:50,557] Trial 66 finished with value: 0.68542659277
36384 and parameters: {'n_estimators': 119, 'max_depth': 21, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:53,030] Trial 67 finished with value: 0.67022316233
81121 and parameters: {'n_estimators': 121, 'max_depth': 11, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:47:57,618] Trial 68 finished with value: 0.67387819549
01397 and parameters: {'n_estimators': 152, 'max_depth': 24, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:00,124] Trial 69 finished with value: 0.62683635920
25077 and parameters: {'n_estimators': 75, 'max_depth': 30, 'min_sample
s_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:03,411] Trial 70 finished with value: 0.68565565507
0338 and parameters: {'n_estimators': 139, 'max_depth': 14, 'min_sample
s_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:05,985] Trial 71 finished with value: 0.68872348480
44812 and parameters: {'n_estimators': 92, 'max_depth': 18, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:08,853] Trial 72 finished with value: 0.68923043623
23524 and parameters: {'n_estimators': 109, 'max_depth': 16, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:09,781] Trial 73 finished with value: 0.65724267579
17427 and parameters: {'n_estimators': 102, 'max_depth': 3, 'min_sample
s_split': 2}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:13,562] Trial 74 finished with value: 0.69050499189
27524 and parameters: {'n_estimators': 128, 'max_depth': 19, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:17,783] Trial 75 finished with value: 0.68847623118
40558 and parameters: {'n_estimators': 131, 'max_depth': 20, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:20,619] Trial 76 finished with value: 0.68276177440
47016 and parameters: {'n_estimators': 125, 'max_depth': 13, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:25,325] Trial 77 finished with value: 0.68971263075
20576 and parameters: {'n_estimators': 169, 'max_depth': 19, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:30,985] Trial 78 finished with value: 0.66150759155
13654 and parameters: {'n_estimators': 185, 'max_depth': 26, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:35,379] Trial 79 finished with value: 0.67782445758
24919 and parameters: {'n_estimators': 144, 'max_depth': 23, 'min_sampl
es_split': 3}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:40,062] Trial 80 finished with value: 0.68957714306
37095 and parameters: {'n_estimators': 166, 'max_depth': 19, 'min_sampl
```

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es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:44,658] Trial 81 finished with value: 0.69017536533
09628 and parameters: {'n_estimators': 163, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:49,364] Trial 82 finished with value: 0.69024411903
12657 and parameters: {'n_estimators': 170, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:54,161] Trial 83 finished with value: 0.68461369012
04539 and parameters: {'n_estimators': 168, 'max_depth': 21, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:48:58,426] Trial 84 finished with value: 0.69008190758
39867 and parameters: {'n_estimators': 155, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:02,729] Trial 85 finished with value: 0.68794944873
66435 and parameters: {'n_estimators': 155, 'max_depth': 19, 'min_sampl
es_split': 6}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:07,349] Trial 86 finished with value: 0.69017795659
35231 and parameters: {'n_estimators': 165, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:12,652] Trial 87 finished with value: 0.67935343710
83089 and parameters: {'n_estimators': 173, 'max_depth': 23, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:18,403] Trial 88 finished with value: 0.64415208853
89173 and parameters: {'n_estimators': 182, 'max_depth': 30, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:22,766] Trial 89 finished with value: 0.68835743342
84932 and parameters: {'n_estimators': 161, 'max_depth': 18, 'min_sampl
es_split': 6}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:27,863] Trial 90 finished with value: 0.65308565004
30858 and parameters: {'n_estimators': 167, 'max_depth': 28, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:32,281] Trial 91 finished with value: 0.68551187868
99169 and parameters: {'n_estimators': 151, 'max_depth': 21, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:36,880] Trial 92 finished with value: 0.68875550307
22945 and parameters: {'n_estimators': 169, 'max_depth': 18, 'min_sampl
es_split': 4}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:42,633] Trial 93 finished with value: 0.69039064110
06462 and parameters: {'n_estimators': 177, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:48,801] Trial 94 finished with value: 0.67539799370
05778 and parameters: {'n_estimators': 177, 'max_depth': 24, 'min_sampl
es_split': 6}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:49:55,398] Trial 95 finished with value: 0.68226448393
63629 and parameters: {'n_estimators': 194, 'max_depth': 22, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:50:00,497] Trial 96 finished with value: 0.68989166247
73248 and parameters: {'n_estimators': 158, 'max_depth': 19, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:50:04,953] Trial 97 finished with value: 0.68633181690
38595 and parameters: {'n_estimators': 157, 'max_depth': 15, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
```

```
[I 2026-04-14 14:50:11,149] Trial 98 finished with value: 0.66601635615
80453 and parameters: {'n_estimators': 174, 'max_depth': 26, 'min_sampl
es_split': 6}. Best is trial 25 with value: 0.6917356919827584.
[I 2026-04-14 14:50:15,533] Trial 99 finished with value: 0.68844943786
36724 and parameters: {'n_estimators': 162, 'max_depth': 17, 'min_sampl
es_split': 5}. Best is trial 25 with value: 0.6917356919827584.
```

	number	value	datetime_start	datetime_complete	duration	pa
25	25	0.691736	2026-04-14 14:45:45.313099	2026-04-14 14:45:49.990891	0 days 00:00:04.677792	
2	2	0.690574	2026-04-14 14:44:50.993420	2026-04-14 14:44:54.599128	0 days 00:00:03.605708	
74	74	0.690505	2026-04-14 14:48:09.782988	2026-04-14 14:48:13.562496	0 days 00:00:03.779508	
93	93	0.690391	2026-04-14 14:49:36.881259	2026-04-14 14:49:42.633462	0 days 00:00:05.752203	
23	23	0.690294	2026-04-14 14:45:40.002773	2026-04-14 14:45:43.654819	0 days 00:00:03.652046	

In [54]:
```
# Train using optimal parameters
model = RandomForestClassifier(
    class_weight='balanced',
    n_estimators=top_5_f1_df.head(1)['params_n_estimators'].item(),
    max_depth=top_5_f1_df.head(1)['params_max_depth'].item(),
    min_samples_split=top_5_f1_df.head(1)['params_min_samples_split'].
    random_state=random_state
)
model.fit(X_train, y_train)
```

Out[54]:
```
▾ RandomForestClassifier
▶ Parameters
```

In [55]:
```
y_pred = model.predict(X_test)
```

In [56]:
```
print(classification_report(y_true=y_test, y_pred=y_pred))
```

```
              precision    recall  f1-score   support

           0       0.98      0.95      0.96     10355
           1       0.62      0.79      0.69      1035

    accuracy                           0.94     11390
   macro avg       0.80      0.87      0.83     11390
weighted avg       0.95      0.94      0.94     11390
```

In [57]:
```
f1_model_report = classification_report(y_true=y_test, y_pred=y_pred,
```

Compare Models

In [ ]:
```
# 2. Convert and transpose
recall_report_df = pd.DataFrame(recall_model_report_df).transpose()
f1_report_df = pd.DataFrame(f1_model_report).transpose()

# 3. Combine multiple reports (e.g., adding a 'Model' column for ident
combined_df = pd.concat([recall_report_df, f1_report_df], keys=['RF op
```

In [64]:
```
combined_df
```

Out[64]:

		precision	recall	f1-score	support
RF optimized for Recall	**0**	0.983590	0.931917	0.957056	10355.000000
	1	0.553515	0.844444	0.668707	1035.000000
	accuracy	0.923968	0.923968	0.923968	0.923968
	macro avg	0.768552	0.888181	0.812882	11390.000000
	weighted avg	0.944509	0.923968	0.930854	11390.000000
RF optimized for F1	**0**	0.978542	0.951231	0.964693	10355.000000
	1	0.618580	0.791304	0.694362	1035.000000
	accuracy	0.936699	0.936699	0.936699	0.936699
	macro avg	0.798561	0.871268	0.829528	11390.000000
	weighted avg	0.945832	0.936699	0.940128	11390.000000

In [65]:
```
model.feature_importances_
```

Out[65]:
```
array([6.62856785e-02, 6.05995047e-02, 9.03900169e-02, 1.01445840e-0
1,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       0.00000000e+00, 0.00000000e+00, 0.00000000e+00, 0.00000000e+0
0,
       2.60059656e-04, 1.78374884e-03, 2.60533729e-04, 2.09601415e-0
4,
       1.49968033e-03, 2.93575262e-04, 4.40192161e-04, 3.55538205e-0
4,
       4.22583210e-03, 8.52128064e-04, 3.51972169e-04, 3.02110751e-0
4,
       9.16247728e-04, 3.06642333e-04, 3.21460516e-04, 1.69076506e-0
4,
       1.24906420e-03, 2.91538203e-04, 2.06885534e-04, 1.35531697e-0
3,
       2.81877643e-04, 3.16866904e-04, 3.73858263e-04, 4.76218125e-0
3,
       8.74444662e-04, 2.62872159e-04, 2.77801866e-04, 1.32006387e-0
3,
       2.70280515e-04, 5.92506764e-04, 3.84747851e-01, 1.02914776e-0
3,
       1.87227051e-02, 1.01112937e-01, 6.69776568e-02, 1.84186719e-0
2,
       9.61421113e-03, 3.74866401e-02, 1.81851808e-02])
```

In [73]:
```
pd.set_option('display.max_rows', None)
```

In [74]:
```
importances = model.feature_importances_
feature_names = df.drop(columns='Is Laundering').columns
feature_imp_df = pd.DataFrame({'Feature': feature_names, 'Gini Importa
    'Gini Importance', ascending=False)
feature_imp_df
```

Out[74]:

	Feature	Gini Importance
94	Payment Format_ACH	0.384748
3	Amount_Paid_USD	0.101446
97	Payment Format_Cheque	0.101113
2	Amount_Received_USD	0.090390
98	Payment Format_Credit Card	0.066978
0	Amount Received	0.066286
1	Amount Paid	0.060600
101	Account_Same	0.037487
96	Payment Format_Cash	0.018723
99	Payment Format_Reinvestment	0.018419
102	Bank_Same	0.018185
100	Payment Format_Wire	0.009614
87	Payment Currency_Saudi Riyal	0.004762
72	Receiving Currency_Saudi Riyal	0.004226
65	Receiving Currency_Bitcoin	0.001784
68	Receiving Currency_Euro	0.001500
83	Payment Currency_Euro	0.001355
91	Payment Currency_US Dollar	0.001320
80	Payment Currency_Bitcoin	0.001249
95	Payment Format_Bitcoin	0.001029
76	Receiving Currency_US Dollar	0.000916
88	Payment Currency_Shekel	0.000874
73	Receiving Currency_Shekel	0.000852
93	Payment Currency_Yuan	0.000593
70	Receiving Currency_Ruble	0.000440

86	Payment Currency_Rupee	0.000374
71	Receiving Currency_Rupee	0.000356
74	Receiving Currency_Swiss Franc	0.000352
78	Receiving Currency_Yuan	0.000321
85	Payment Currency_Ruble	0.000317
77	Receiving Currency_Yen	0.000307
75	Receiving Currency_UK Pound	0.000302
69	Receiving Currency_Mexican Peso	0.000294
81	Payment Currency_Brazil Real	0.000292
84	Payment Currency_Mexican Peso	0.000282
90	Payment Currency_UK Pound	0.000278
92	Payment Currency_Yen	0.000270
89	Payment Currency_Swiss Franc	0.000263
66	Receiving Currency_Brazil Real	0.000261
64	Receiving Currency_Australian Dollar	0.000260
67	Receiving Currency_Canadian Dollar	0.000210
82	Payment Currency_Canadian Dollar	0.000207
79	Payment Currency_Australian Dollar	0.000169
38	UK Pound_Amount_Received_Mean	0.000000
37	UK Pound_Amount_Paid_Median	0.000000
36	UK Pound_Amount_Paid_Mean	0.000000
31	Yen_Amount_Received_Median	0.000000
30	Yen_Amount_Received_Mean	0.000000
29	Yen_Amount_Paid_Median	0.000000
28	Yen_Amount_Paid_Mean	0.000000
27	Rupee_Amount_Received_Median	0.000000
26	Rupee_Amount_Received_Mean	0.000000
25	Rupee_Amount_Paid_Median	0.000000
24	Rupee_Amount_Paid_Mean	0.000000
23	Yuan_Amount_Received_Median	0.000000
22	Yuan_Amount_Received_Mean	0.000000

21	Yuan_Amount_Paid_Median	0.000000
20	Yuan_Amount_Paid_Mean	0.000000
19	Australian Dollar_Amount_Received_Median	0.000000
18	Australian Dollar_Amount_Received_Mean	0.000000
17	Australian Dollar_Amount_Paid_Median	0.000000
16	Australian Dollar_Amount_Paid_Mean	0.000000
15	Euro_Amount_Received_Median	0.000000
14	Euro_Amount_Received_Mean	0.000000
13	Euro_Amount_Paid_Median	0.000000
12	Euro_Amount_Paid_Mean	0.000000
11	Bitcoin_Amount_Received_Median	0.000000
10	Bitcoin_Amount_Received_Mean	0.000000
9	Bitcoin_Amount_Paid_Median	0.000000
8	Bitcoin_Amount_Paid_Mean	0.000000
7	US Dollar_Amount_Received_Median	0.000000
6	US Dollar_Amount_Received_Mean	0.000000
5	US Dollar_Amount_Paid_Median	0.000000
4	US Dollar_Amount_Paid_Mean	0.000000
35	Mexican Peso_Amount_Received_Median	0.000000
34	Mexican Peso_Amount_Received_Mean	0.000000
33	Mexican Peso_Amount_Paid_Median	0.000000
32	Mexican Peso_Amount_Paid_Mean	0.000000
63	Shekel_Amount_Received_Median	0.000000
55	Brazil Real_Amount_Received_Median	0.000000
56	Saudi Riyal_Amount_Paid_Mean	0.000000
57	Saudi Riyal_Amount_Paid_Median	0.000000
58	Saudi Riyal_Amount_Received_Mean	0.000000
59	Saudi Riyal_Amount_Received_Median	0.000000
60	Shekel_Amount_Paid_Mean	0.000000
61	Shekel_Amount_Paid_Median	0.000000
62	Shekel_Amount_Received_Mean	0.000000

46	Canadian Dollar_Amount_Received_Mean	0.000000
45	Canadian Dollar_Amount_Paid_Median	0.000000
44	Canadian Dollar_Amount_Paid_Mean	0.000000
43	Ruble_Amount_Received_Median	0.000000
42	Ruble_Amount_Received_Mean	0.000000
41	Ruble_Amount_Paid_Median	0.000000
40	Ruble_Amount_Paid_Mean	0.000000
47	Canadian Dollar_Amount_Received_Median	0.000000
39	UK Pound_Amount_Received_Median	0.000000
48	Swiss Franc_Amount_Paid_Mean	0.000000
49	Swiss Franc_Amount_Paid_Median	0.000000
50	Swiss Franc_Amount_Received_Mean	0.000000
51	Swiss Franc_Amount_Received_Median	0.000000
52	Brazil Real_Amount_Paid_Mean	0.000000
53	Brazil Real_Amount_Paid_Median	0.000000
54	Brazil Real_Amount_Received_Mean	0.000000

In [ ]:

4. Storytelling With Data plot

Reproduce any graph of your choice in chapter seven (p. 165-185) of the Storytelling With Data book as best you can. You do not have to get the exact data values right, just the overall look and feel.

In [ ]:

```
# Figure 8.3 (p. 191)
import seaborn as sns
import matplotlib.ticker as tkr

products = [
    'Product A',
    'Product A',
    'Product A',
    'Product A',
    'Product A',
    'Product A',
    'Product A',
    'Product B',
```

```
    'Product B',
    'Product B',
    'Product B',
    'Product B',
    'Product B',
    'Product B',
    'Product C',
    'Product C',
    'Product C',
    'Product C',
    'Product C',
    'Product C',
    'Product C',
    'Product D',
    'Product D',
    'Product D',
    'Product D',
    'Product D',
    'Product D',
    'Product D',
    'Product E',
    'Product E',
    'Product E',
    'Product E',
    'Product E',
    'Product E',
    'Product E',
]
years = [
    # A
    '2008',
    '2009',
    '2010',
    '2011',
    '2012',
    '2013',
    '2014',
    # B
    '2008',
    '2009',
    '2010',
    '2011',
    '2012',
    '2013',
    '2014',
    # C
    '2008',
    '2009',
    '2010',
    '2011',
    '2012',
    '2013',
```

```
    '2014',
    # D
    '2008',
    '2009',
    '2010',
    '2011',
    '2012',
    '2013',
    '2014',
    # E
    '2008',
    '2009',
    '2010',
    '2011',
    '2012',
    '2013',
    '2014',
]
prices = [
    # A
    390,
    410,
    415,
    385,
    300,
    285,
    280,
    # B
    390,
    410,
    415,
    385,
    300,
    285,
    280,
    # C
    0,
    0,
    100,
    200,
    205,
    230,
    185,
    # D
    0,
    0,
    150,
    210,
    202,
    200,
    199,
    # E
```

```
    0,
    0,
    0,
    0,
    0,
    100,
    205,
]
df = pd.DataFrame({
    'Product': products,
    'Year': years,
    'Price': prices
})

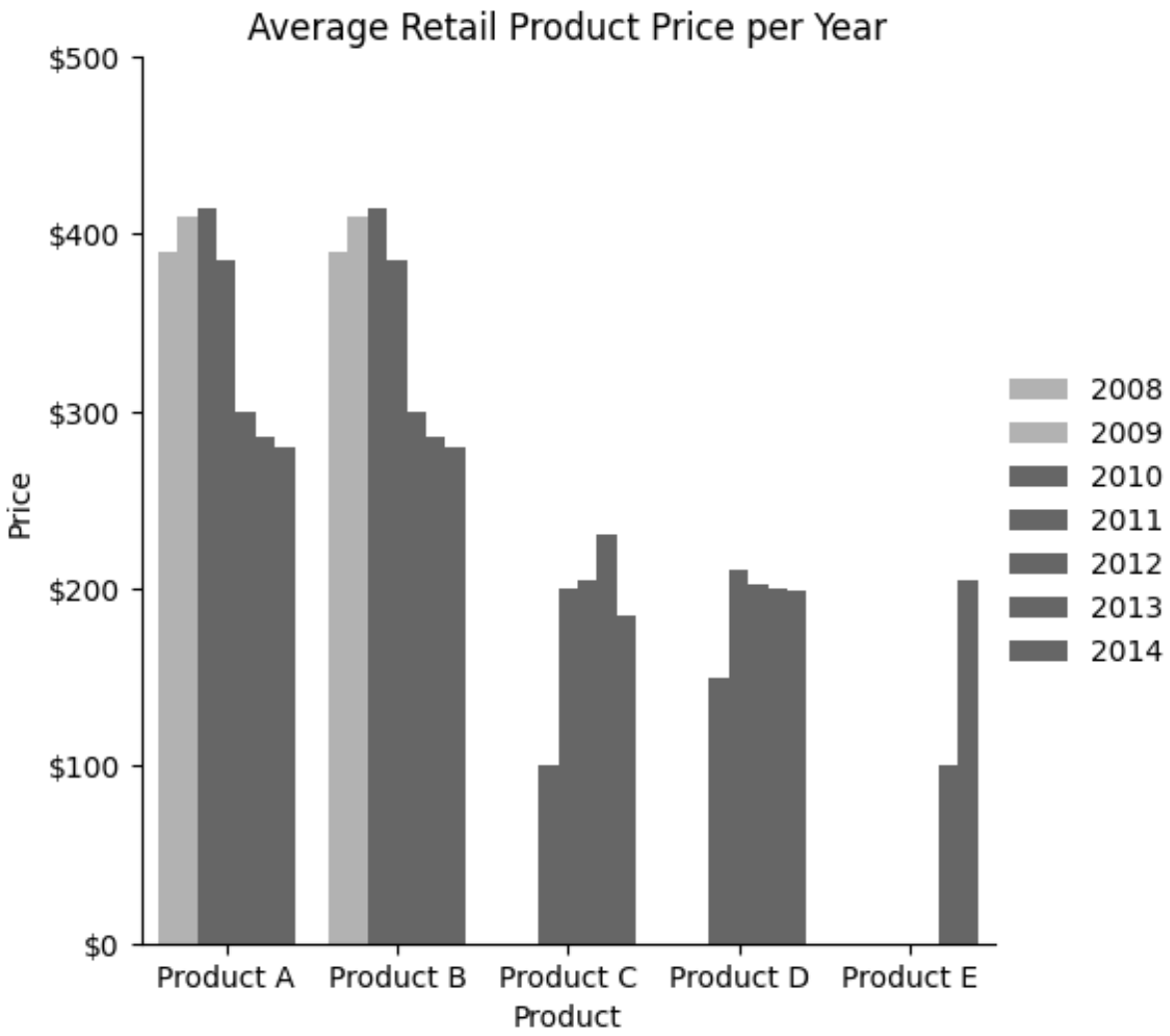
# Draw a nested barplot by species and sex
colors = ["gray", "gray", "black", "black", "black", "black", "black"]

g = sns.catplot(
    data=df, kind="bar",
    x="Product", y="Price", hue="Year",
    palette=colors, alpha=.6,
)
# Apply dollar sign formatting
fmt = '${x:,.0f}' # Adds $ and comma separators
tick = tkr.StrMethodFormatter(fmt)
for ax in g.axes.flat:
    ax.yaxis.set_major_formatter(tick)

plt.ylim(None, 500) # Sets y-max to 100

g.legend.set_title("")
plt.title("Average Retail Product Price per Year")
```

Out[ ]: Text(0.5, 1.0, 'Average Retail Product Price per Year')


Average Retail Product Price per Year
$500
$400
$300
$200
$100
$0
Price
Product A
Product B
Product C
Product D
Product E
Product
2008
2009
2010
2011
2012
2013
2014